CITS4404 Project — Building AI Trading Bots: Nature-Inspired Optimisation Algorithms Implementation and Experimentation, and Evaluation

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4646 words

I. ALGORITHMS INVESTIGATED

Six optimisation algorithms were evaluated across all experimental configurations, spanning four structural categories: swarm-based, physics-inspired, ecosystem-inspired, and classical local search.

Particle Swarm Optimization, originally proposed by Kennedy and Eberhart [1] and extended with an inertia weight by Shi and Eberhart [2], is a swarm method in which each particle maintains both a personal best position and a shared global best. The inertia weight w governs the balance between exploration and exploitation through the velocity update rule. Atomic Orbital Search (AOS) [3] is a physics-inspired method detailed in Section III; it employs shell-based clustering around a nucleus (global best) as its primary exploration mechanism, without per-agent personal memory. African Buffalo Optimization (ABO) [4] encodes two herd vocalisations — an exploratory foraging call and an exploitative regrouping call — and like PSO retains both personal and global best positions. Manta Ray Foraging Optimization (MRFO) [5] models three foraging behaviours (chain, cyclone, and somersault foraging) but maintains only a global best, with no per-agent memory. Symbiotic Organisms Search (SOS) [6] simulates mutualistic, commensal, and parasitic ecological interactions, applying all three phases to every organism per iteration; this yields three fitness evaluations per organism per iteration and requires no algorithm-specific hyperparameters. Generalized Pattern Search (GPS) [7] is a classical single-state local optimiser that evaluates a fixed pattern of directions from the current best solution, with no population and no historical memory beyond the current position; it serves as a deliberate baseline to quantify the value of population-based diversity.

Table I summarises the distinguishing characteristics of each algorithm.

TABLE I. Summary of the six optimisation algorithms evaluated. Memory column describes which historical positions are retained per agent. Eval/iter denotes fitness evaluations per agent per iteration.

Algorithm	Year	Category	Population-based	Memory	Eval/iter
AOS	2021	Physics	Yes	Global (nucleus)	1
PSO	1998	Swarm	Yes	Personal + Global	1
MRFO	2020	Swarm	Yes	Global only	1
SOS	2014	Ecosystem	Yes	Global only	3
ABO	2015	Swarm	Yes	Personal + Global	1
GPS	—	Local search	No	Current best only	1

II. BOT DESIGN AND PARAMETRISATION

A. Trading Signal and Fitness

Each strategy produces a short-window (low-period) and a long-window (high-period) smoothed price series. The trading signal detects crossover events in their difference using a two-point weighted kernel:

$$d_t = \text{Series}_{\text{short},t} - \text{Series}_{\text{long},t} \quad , \quad C_t = \tfrac{1}{2}(\text{sign}(d_t) - \text{sign}(d_{t-1})) \tag{1}$$

$$S = \begin{cases} \text{buy} & \text{if } C_t = -1 \text{ (short crosses below long)} \\ \text{sell} & \text{if } C_t = +1 \text{ (short crosses above long)} \end{cases} \tag{2}$$

where d_t is the difference of the two smoothed series at time t, $C_t \in \{-1, 0, +1\}$ is the crossover indicator, and S is the resulting trade action. The signal is non-zero only at the moment of a crossover; it is zero on all other timesteps.

The fitness function measures final USD holdings after simulating trades over the full training price series:

$$\text{Fitness}(X) = \text{USD}_{\text{final}} \tag{3}$$

where X is the candidate parameter vector, $\text{USD}_{\text{final}}$ is the cash held at the end of the simulation, and the starting balance is $\text{USD}_{\text{initial}} = \$1000$. A transaction fee of 3% is applied to each trade. The simulation is a single continuous pass over the training data with no mid-run resets.

One acknowledged limitation is that the fitness function imposes no penalty for non-trading: if no crossover ever occurs, $C_t = 0$ for all t and the bot holds cash throughout, returning $\text{Fitness}(X) = \text{USD}_{\text{initial}}$. This is a valid but uninformative fixed point. The bad/broken optimum was observed empirically — MRFO converged to parameter regions with no crossover signal under certain configurations. A risk-adjusted metric such as the Sharpe ratio would penalise this behaviour but was not adopted here; the limitation is documented rather than silently corrected.

B. Strategy Configurations

Five strategy configurations define the hypothesis space presented to each optimiser.

Strategy 1 — Double SMA crossover ($d = 2$). Both the short window $x_0 \in [2, 50]$ and the long window $x_1 \in [51, 200]$ [8] are optimised. The non-overlapping constraint (short < long) is guaranteed by construction through the parameter bounds, with no runtime enforcement required. The SMA assigns uniform weight to all prices within the window, making it insensitive to the ordering of recent versus older prices.

Strategy 2 — Double LMA crossover ($d = 2$). The parameter structure is identical to Strategy 1 ($x_0 \in [2, 50]$, $x_1 \in [51, 200]$). The LMA assigns linearly decaying weights so that more recent prices contribute more than older ones, providing momentum sensitivity that the uniform SMA cannot capture. The optimiser thus faces the same two-dimensional search space as Strategy 1 but with a smoother series that responds differently to regime changes.

Strategy 3 — EMA with shared α ($d = 3$). A short span, a long span, and a single shared smoothing parameter $\alpha \in (0, 1)$ are optimised jointly. Sharing one α value across both EMAs constrains both series to decay at identical rates; the frequency contrast that drives the crossover signal therefore depends entirely on the span difference rather than on independent decay dynamics. This effectively collapses the degree of freedom to approximately 2.5 effective dimensions in practice.

Strategy 4 — EMA with independent α ($d = 4$). Short span, long span, α_{short}, and α_{long} are each optimised independently, along with additional EMA window parameters. Decoupling the smoothing parameters gives the short-term EMA its own reactivity and the long-term EMA its own stability, restoring the frequency contrast that shared α eliminates. The expanded search space ($d = 4$ versus $d = 3$) carries a higher exploration cost but provides genuinely greater expressiveness: the two EMAs can now operate at qualitatively different timescales simultaneously. Empirically, this configuration produced the highest Kruskal-Wallis H-statistics across all strategy conditions, suggesting that the larger search space amplifies inter-algorithm performance differences rather than masking them.

Strategy 5 — Weighted MA combination ($d = 14$). SMA, LMA, and EMA signals are combined as a weighted sum with learned weights w_1, w_2, w_3. All three weights are treated as free optimisation dimensions and normalised at evaluation time (each w_i divided by $w_1 + w_2 + w_3$), yielding a 14-dimensional parameter vector that includes window parameters for each MA type plus EMA spans and smoothing factors. Deriving $w_3 = 1 - w_1 - w_2$, as the project brief implies, would create asymmetric optimisation pressure because w_3 is never directly searched; retaining all three as free variables ensures each weight receives equal search effort. The 14-dimensional space is the largest tested, and the curse of dimensionality is a genuine concern when each optimiser is limited to 50 runs.

C. Justified Deviations from Project Brief

TABLE II. Summary of implementation decisions that deviate from the project brief, with justification for each.

Deviation	Brief approach	Our implementation	Justification
EMA normalisation	Optional	Enabled by default	Unnormalised kernels produce span-dependent amplitude distortion, making fitness values non-comparable across parameter configurations
$d = 14$ weight derivation	w_3 derived as $1 - w_1 - w_2$	All three weights free; normalised at eval time	Derived w_3 receives no direct search pressure, creating asymmetric optimisation bias
n_{shells} assignment	Fixed shell count	$\min(n_{\text{shells}}, \lvert\text{remaining}\rvert)$	Prevents degenerate empty shell states in high-dimensional runs where population may be smaller than the configured shell count
EMA α across spans	Shared α (Strategy 3)	Independent α_{short}, α_{long} (Strategy 4)	Shared α constrains both series to identical decay rates, eliminating the frequency contrast that makes the crossover signal informative

III. ALGORITHM SELECTION

All six algorithms described in Section I were evaluated. This section explains the rationale behind each inclusion and details the primary algorithm, AOS.

GPS as a deliberate baseline. The central empirical question is whether population-based global search is necessary on a moving-average (MA) parameter fitness landscape, or whether a single-state local optimiser is sufficient. GPS [7] has no population, no escape mechanism from local optima, and no memory of previously evaluated positions beyond the current best. Including it provides a direct lower bound: if GPS performs competitively, population diversity contributes little; if GPS underperforms consistently, the landscape demands it.

PSO as a population-based benchmark. PSO [2] is one of the most extensively studied metaheuristics and provides a reliable population-based performance reference. It is also structurally related to AOS — both use a global nucleus/best to guide the population — making a paired comparison directly informative about the contribution of personal best memory. If AOS and PSO are statistically indistinguishable, per-particle historical memory may or may not add marginal benefit in this problem domain.

AOS as the physics-based algorithm compared to the others being biology-based. The MA parameter fitness landscape is expected to be multimodal: distinct parameter combinations can produce similar short-term profits through qualitatively different mechanisms, creating multiple local peaks of comparable height. AOS [3] was selected because its shell-based population structure explicitly supports simultaneous exploitation of multiple landscape regions. Candidate solutions (electrons) are assigned to concentric shells around the nucleus (current global best) according to their fitness rank; electrons in inner shells exploit promising regions while outer-shell electrons conduct broader exploration. Transitions between shells are governed by a quantum-mechanical photon absorption and emission analogy: an electron absorbs energy (moves to an outer shell, exploration) when its transition energy falls below the shell's binding energy, and emits energy (moves to an inner shell, exploitation) otherwise. This structured push-pull between shells provides more persistent population spread than PSO's convergent swarm behaviour, which collapses toward the global best over time. AOS was published in 2021 and has not, to our knowledge, been applied to financial parameter optimisation in the literature.

The pseudocode below summarises the core AOS iteration:

```
Initialise N electrons randomly in the search space
Set nucleus := position of the best electron (highest fitness)
Assign electrons to K shells by fitness rank (binding energy)

FOR each iteration t = 1, 2, ..., T:
    FOR each shell k = 1, ..., K:
        Compute shell binding energy E_k
        FOR each electron i in shell k:
            Sample transition energy E
            IF E < E_k  (electron absorbs energy):
                Move to outer shell; apply exploration step
            ELSE  (electron emits energy):
                Move to inner shell; apply exploitation step
```

```
            Update position via quantum transition equation
            Evaluate fitness of new position
    Apply LE perturbation: randomly displace the
        lowest-energy electron to escape local optima
    Update nucleus if any electron improves best fitness

RETURN nucleus position as best solution found
```

LE degradation — an observed weakness. The Lowest Energy (LE) perturbation step is designed to prevent stagnation by randomly displacing the electron with the worst fitness. In practice, the LE designation is not stable: as other electrons improve their fitness across iterations, the electron currently labelled LE may no longer occupy the position most in need of escape relative to the updated nucleus. The original paper [3] does not address this drift. This was observed empirically across runs and is logged per iteration; it is documented here rather than patched, as any correction would constitute a modification to the published algorithm.

Metaparameter choices. Population size was scaled with dimensionality — larger populations were used for the $d = 14$ weighted strategy to partially offset the curse of dimensionality. The shell count n_{shells} is implemented as an upper cap rather than a fixed assignment: each iteration uses $\min(n_{\text{shells}}, |\text{remaining population}|)$ shells to prevent degenerate empty-shell states that can arise in high-dimensional runs where effective population diversity is reduced. The iteration budget was fixed uniformly across all six algorithms to ensure a controlled comparison of per-iteration effectiveness.

IV. EXPERIMENTS AND EVALUATION

A. Dataset

Daily BTC/USD OHLCV data sourced from Kaggle spanning 2014 to 2022 was used. The training set covers the pre-2020 period (approximately 2014–2019) and the test set covers 2020–2022. The split was not chosen arbitrarily: post-2020 data encompasses the COVID-19 market crash of March 2020, the 2021 bull run, and the 2022 bear market, constituting three qualitatively distinct market regimes. This is an intentional out-of-distribution stress test rather than a random holdout. Its purpose is to measure whether parameter configurations optimised on one regime generalise when the underlying market dynamics shift substantially — a more meaningful evaluation of strategy robustness than a matched-regime random split would provide.

B. Experimental Design

The full factorial design comprised 6 algorithms $\times$ 5 strategies $\times$ 2 initialisation modes $\times$ 50 runs $=$ 3000 total trials. Two initialisation modes were tested: fixed initialisation, in which all runs begin from a deterministic seed position, and random initialisation, in which each run begins from a position drawn uniformly at random from the search space. The paired comparison of fixed versus random initialisation tests sensitivity to starting conditions. Strong dependence on initialisation would indicate algorithmic fragility, with convergence driven by the seed rather than by the search mechanism. A run count of 50 was chosen as a practical compromise between statistical power and the total computational cost of 3000 trials; it is sufficient to apply non-parametric distributional tests but modest relative to what would be preferred for high-variance optimisers.

C. Statistical Analysis

The Kruskal-Wallis H-test was used in preference to one-way ANOVA because the structural properties of optimiser output systematically violate the normality assumption required by ANOVA. Optimisers can produce runs that converge to the broken no-trade optimum (fitness near \$1000, the starting balance), occasional jackpot runs that far exceed the typical distribution, and heavy-tailed spread in between. These outliers are not noise — they reflect real behavioural modes of the algorithms — and would distort ANOVA F-statistics. Kruskal-Wallis operates on ranks and is robust to such distributional shapes.

For each of the ten strategy $\times$ initialisation mode combinations, the Kruskal-Wallis test was applied with algorithm as the factor variable, holding strategy and initialisation constant. This isolates the algorithm effect from strategy-level confounds: testing across all strategies simultaneously would conflate the variance introduced by different fitness landscape shapes with the variance attributable to algorithmic differences. Pairwise post-hoc comparisons used Dunn's test with Bonferroni correction. A significance threshold of $\alpha = 0.05$ was applied throughout.

V. RESULTS AND ANALYSIS

A. Training Results

FIG. 1. Dunn's post-hoc pairwise p-values (Bonferroni-corrected) for all strategy $\times$ initialisation combinations — training fitness. Red indicates significant difference ($p < 0.05$); green indicates no significant difference.

The Kruskal–Wallis test was significant across all ten strategy $\times$ initialisation combinations on training fitness ($p < 10^{-34}$ in every case; Table III, training columns). Algorithm choice has a universally significant effect on training fitness. The `ema_independent` strategy ($d = 4$) yielded the highest H-statistics ($H = 223.17$ fixed, 244.91 random), exceeding the $d = 14$ `weighted` strategy ($H = 213.19$, 224.80). This suggests that decoupling the two

TABLE III. Kruskal–Wallis H-test results comparing optimisation algorithms for each strategy and initialisation mode, on *training* and *test* fitness. On training fitness, algorithm choice is universally significant ($p < 10^{-34}$). On test fitness the effect is strategy-dependent: `ema_shared` loses all significance ($p > 0.90$), while `ema_independent` retains the strongest effect ($H > 179$).

		Training		**Test**	
Strategy	**Start**	H	p	H	p
sma	fixed	200.63	2.1×10^{-41}	46.59	6.9×10^{-9}
sma	random	198.09	7.3×10^{-41}	85.68	5.4×10^{-17}
lma	fixed	176.56	2.9×10^{-36}	73.95	1.5×10^{-14}
lma	random	170.28	6.3×10^{-35}	58.34	2.7×10^{-11}
ema_shared	fixed	206.76	1.0×10^{-42}	0.60	9.9×10^{-1}
ema_shared	random	190.41	3.2×10^{-39}	1.51	9.1×10^{-1}
ema_independent	fixed	223.17	3.1×10^{-46}	184.26	6.6×10^{-38}
ema_independent	random	244.91	6.8×10^{-51}	179.13	8.2×10^{-37}
weighted	fixed	213.19	4.3×10^{-44}	78.91	1.4×10^{-15}
weighted	random	224.80	1.4×10^{-46}	78.54	1.7×10^{-15}

EMA smoothing parameters amplifies inter-algorithm performance differences more than further increasing raw dimensionality: algorithms with stronger exploration mechanisms appear to extract more from the qualitative change in expressiveness than from additional degrees of freedom alone.

Three patterns emerge from Dunn's pairwise post-hoc comparisons on training fitness (Figure 1). First, GPS is significantly separated from the population-based methods in the large majority of training conditions, consistent with single-state local search struggling to navigate the multimodal MA fitness landscape: without population diversity, GPS converges to the nearest local optimum from its starting position. The clearest exception is `ema_independent`, where GPS and MRFO are not significantly different ($p = 0.16$ fixed, $p = 1.00$ random); on this strategy MRFO collapses to the no-trade fixed point ($1,000, the starting balance) in every fixed-initialisation run (median = $1,000, CV = 0), so the non-significance reflects two algorithms failing to the same floor rather than performing comparably well.

Second, the apparent PSO–SOS–AOS equivalence is strategy-dependent rather than universal. On `weighted` and on the broken $d = 2$ strategies (`sma`, `lma`), PSO, SOS, and AOS are largely statistically indistinguishable under Bonferroni-corrected Dunn's comparisons (non-significance under Bonferroni is not formal equivalence; we report what the test shows). On `ema_independent`, however, this cluster breaks apart: AOS separates significantly from PSO ($p = 1.7 \times 10^{-3}$ fixed, $p = 2.5 \times 10^{-6}$ random) and from SOS. The most expressive landscape is therefore the one on which the optimisers are most cleanly differentiated, and on which AOS no longer tracks the memory-bearing methods — a divergence examined further in the test results below.

Third, SOS exhibits zero within-cell variance on the broken $d = 2$ strategies: identical training fitness across all 50 `sma` runs and all 50 `lma` runs (CV = 0.000 in both cases; Table IV). This is consistent with its three-phase update collapsing population diversity within a small number of iterations. SOS also requires three fitness evaluations per organism per iteration; its mean wall-clock runtime (14,730 ms) is roughly four times that of PSO (3,706 ms), so a compute-normalised comparison would materially disadvantage SOS relative to PSO. MRFO and ABO occupy intermediate positions that vary by strategy, with no consistent dominance pattern relative to the PSO–SOS–AOS cluster.

B. Test Results — Performance Against the Passive Baseline

A passive buy-and-hold strategy over the 2020–2022 test period, with matched 3% entry and exit fees, produced $5,660.26 from a $1,000 starting balance. Of the 3,000 optimised bots, only 62 (2.07%) strictly exceeded this baseline; no algorithm × strategy median exceeded it under any condition (Figure 2). Win rates were heavily stratified by strategy (Table V). Each strategy spans 6 algorithms × 2 initialisation modes × 50 trials = 600 runs. `sma` and `lma` (both $d = 2$) yielded zero wins across their 1,200 combined runs; `ema_independent` ($d = 4$) produced only 2/600 (0.33%), both under fixed initialisation; `weighted` ($d = 14$) produced 26/600 (4.33%); and `ema_shared` ($d = 3$) produced 34/600 (5.67%). The wins are thus almost entirely confined to two strategies — and, as Section V (Why Training Fitness Did Not Predict Test Fitness) shows, those are precisely the two strategies on which training fitness carries the least information about test fitness.

Variance within the two winning strategies differs substantially (Table VI). `weighted`-strategy bots converge tightly — every algorithm except GPS has a test CV below 0.26, and SOS reaches CV = 0.107 around a median of $4,750.

TABLE IV. Summary of *training* fitness by algorithm and strategy, pooled across both initialisation modes (100 runs per cell). Note the very high training fitness reached on `ema_shared` (medians up to $133,000), which Section V shows does not transfer to test.

Algo	**Strategy**	**Mean**	**Std**	**CV**	**Median**	**Max**
abo	ema_independent	10,750.12	1,802.05	0.168	10,347.23	15,352.63
aos	ema_independent	19,631.96	5,008.31	0.255	21,904.21	21,904.21
gps	ema_independent	5,179.20	3,858.59	0.745	4,543.98	12,939.69
mrfo	ema_independent	1,000.00	0.00	0.000	1,000.00	1,000.00
pso	ema_independent	12,092.59	1,255.72	0.104	11,463.70	15,303.77
sos	ema_independent	13,273.95	799.34	0.060	13,197.07	15,607.47
abo	ema_shared	93,661.07	24,627.52	0.263	90,428.45	157,746.47
aos	ema_shared	67,128.07	26,684.54	0.398	60,697.82	229,103.39
gps	ema_shared	11,989.89	14,016.59	1.169	7,882.87	59,090.09
mrfo	ema_shared	86,615.44	31,961.19	0.369	78,405.64	237,614.46
pso	ema_shared	103,646.40	22,819.48	0.220	102,724.96	193,483.44
sos	ema_shared	132,840.58	30,893.95	0.233	128,243.17	215,646.97
abo	lma	802.79	16.63	0.021	800.11	818.93
aos	lma	805.44	15.38	0.019	809.92	818.93
gps	lma	463.95	120.20	0.259	461.38	818.93
mrfo	lma	810.79	11.45	0.014	818.93	818.93
pso	lma	798.10	16.98	0.021	800.11	818.93
sos	lma	818.93	0.00	0.000	818.93	818.93
abo	sma	930.41	13.21	0.014	932.50	932.50
aos	sma	931.76	0.66	0.001	931.18	932.50
gps	sma	551.42	118.34	0.215	542.81	917.97
mrfo	sma	932.50	0.00	0.000	932.50	932.50
pso	sma	909.75	48.31	0.053	932.50	932.50
sos	sma	932.50	0.00	0.000	932.50	932.50
abo	weighted	11,339.38	1,200.41	0.106	11,385.64	14,205.09
aos	weighted	10,697.16	2,147.99	0.201	10,355.90	29,952.87
gps	weighted	5,124.90	3,544.28	0.692	5,295.31	12,601.00
mrfo	weighted	14,049.10	9,313.82	0.663	11,952.69	65,347.85
pso	weighted	12,744.28	1,112.69	0.087	12,602.90	17,561.96
sos	weighted	13,215.13	549.20	0.042	13,242.64	14,531.01

TABLE V. Runs exceeding the passive buy-and-hold baseline under strict > comparison. The baseline is $5,660.26; a threshold of $5,661 is applied here for reporting clarity. No run falls in the interval ($5,660.26, $5,661], so the integer threshold yields counts identical to the exact baseline. Counts are out of 50 runs per cell; 3,000 trials in total.

	ema_ind		**ema_shared**		**lma**		**sma**		**weighted**	
algo	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix	rnd
abo	0	0	1	3	0	0	0	0	3	6
aos	0	0	1	3	0	0	0	0	2	2
gps	2	0	3	3	0	0	0	0	0	0
mrfo	0	0	3	3	0	0	0	0	0	3
pso	0	0	2	6	0	0	0	0	2	1
sos	0	0	4	2	0	0	0	0	4	3
total	**2**	**0**	**14**	**20**	**0**	**0**	**0**	**0**	**11**	**15**

In contrast, `ema_shared` winners come from highly dispersed distributions: test CV exceeds 0.85 for every algorithm and exceeds 1.0 for GPS, around a median pinned at the no-trade floor of $1,000. This dispersion is the signature of stochastic discovery rather than reliable optimisation: a `weighted` win is a comparatively repeatable outcome, whereas an `ema_shared` win is a draw from a long right tail above an otherwise collapsed distribution.

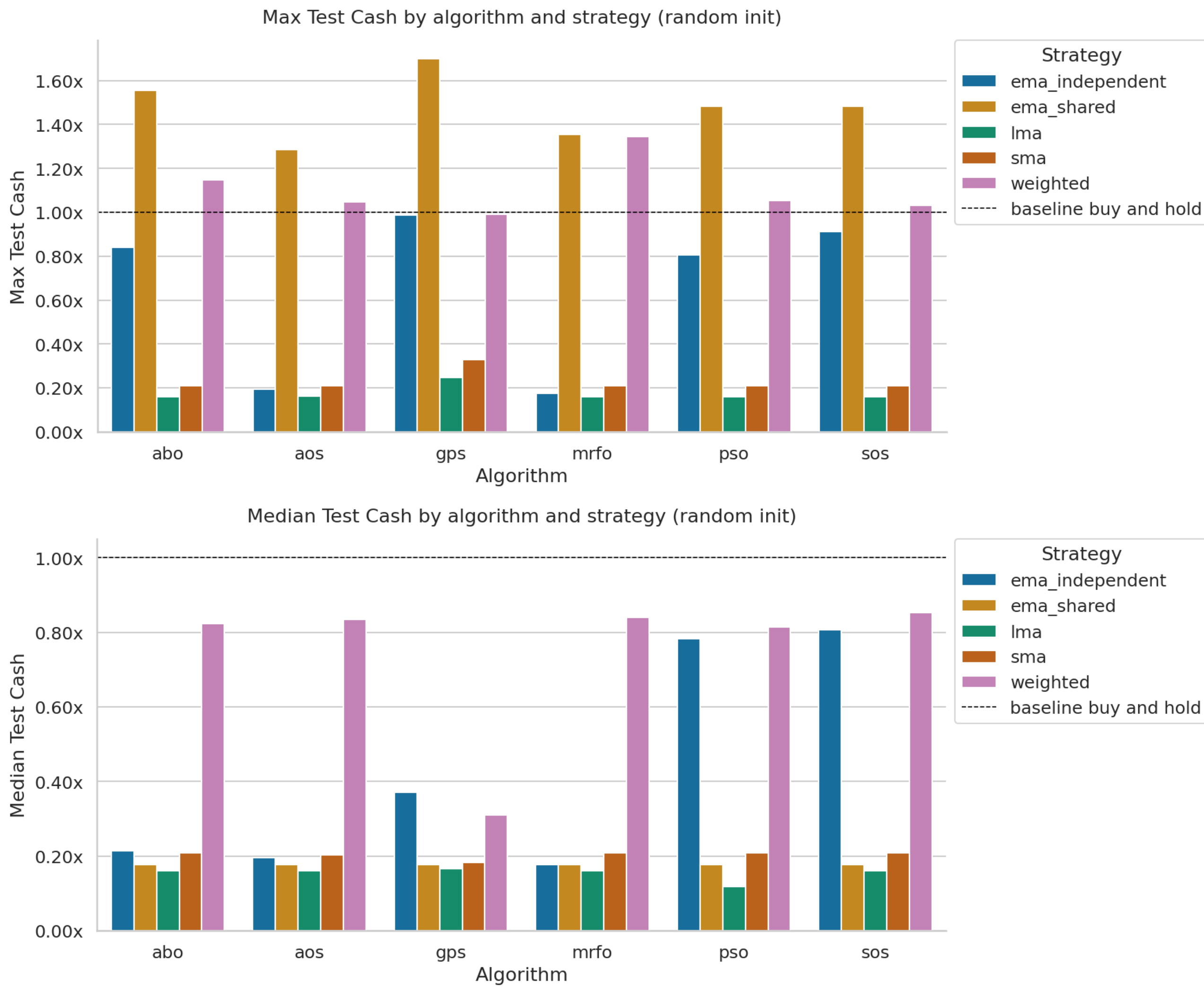

FIG. 2. Median test cash by algorithm and strategy, normalised to the buy-and-hold baseline ($5,660.26 from $1,000 initial under matched 3% entry and exit fees), for fixed (top) and random (bottom) initialisation. No algorithm × strategy median exceeds the baseline (1.00× line) under any condition.

C. Test Results — Statistical Differentiation

On test data, the strong pairwise differences observed during training largely collapse (Figure 4). The most extreme case is `ema_shared`: under both fixed and random initialisation, every pairwise p-value equals 1.000, and the Kruskal–Wallis test is itself non-significant ($H = 0.60$, $p = 0.99$ fixed; $H = 1.51$, $p = 0.91$ random; Table III). All six algorithms produce statistically indistinguishable test fitness distributions on the strategy that produced the most baseline-beating runs — a direct indication that those wins are not attributable to optimiser quality.

The one strategy that retains strong test-side differentiation is `ema_independent` ($H = 184.26$ fixed, 179.13 random). Here the training-stage divergence within the PSO–SOS–AOS cluster persists onto test, but in a specific direction: PSO and SOS retain high test medians (pooled medians $4,435 and $4,566 respectively), whereas AOS collapses to a pooled test median of $1,105 — close to ABO ($1,239) and MRFO ($1,000), and well below GPS ($1,982). AOS trains to a high `ema_independent` fitness but does not retain it out-of-sample. Since the structural difference between AOS and PSO/SOS most relevant here is the absence of per-agent personal-best memory in AOS, this at least suggests that personal-best memory aids generalisation on the most expressive landscape; however, the present design does not isolate memory as a variable, so this remains a hypothesis for targeted future work rather than a demonstrated mechanism.

A caution applies to all surviving test-side significances. A significant rank difference on test fitness, on strategies where a large fraction of runs sit at broken fixed points (no-trade $1,000, and the strategy-specific SMA and LMA

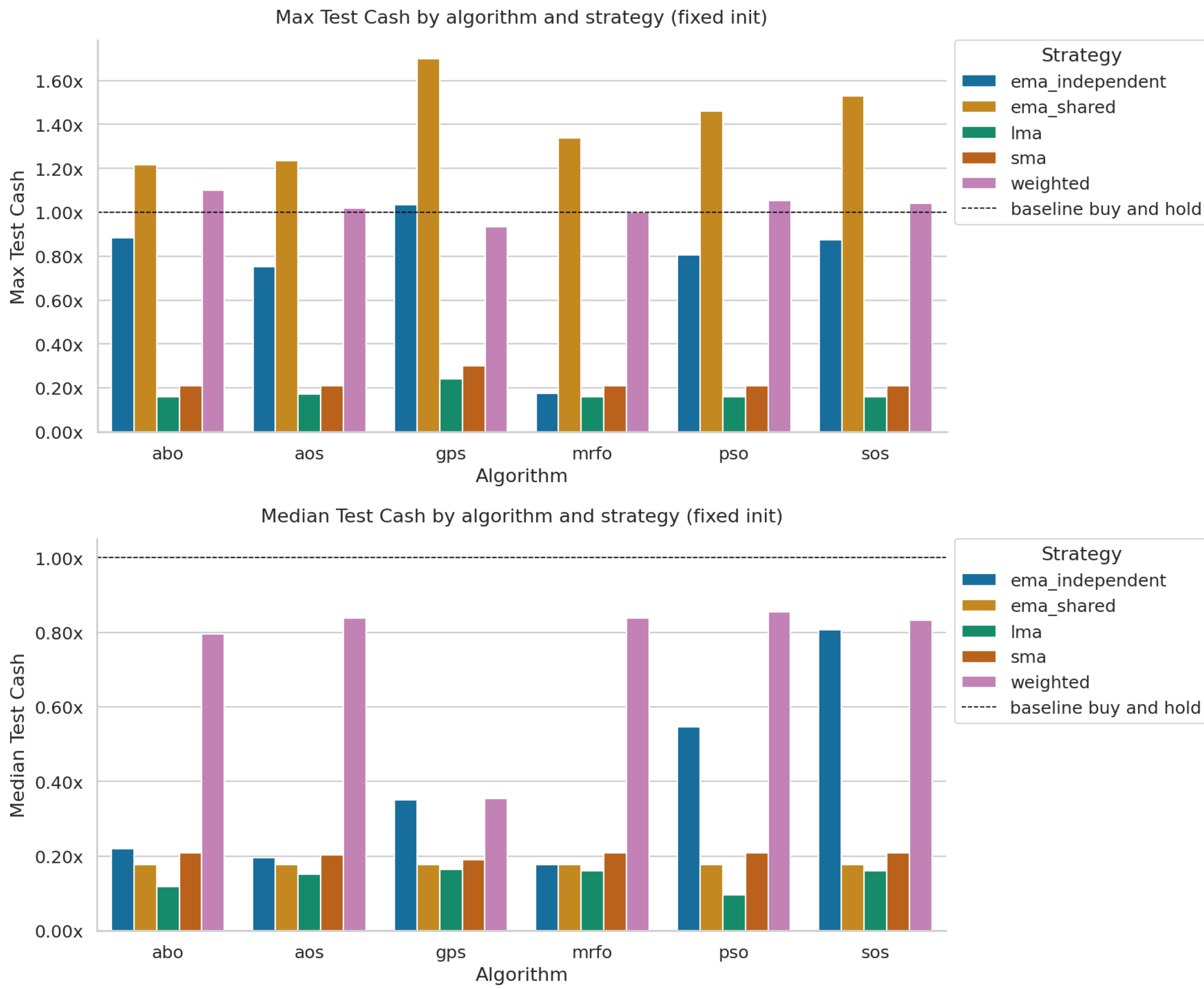

FIG. 3. Maximum test cash by algorithm and strategy, normalised to the buy-and-hold baseline, for fixed (top) and random (bottom) initialisation. Bars above 1.00× indicate at least one run in 50 exceeded passive holding; concentration of such bars in `ema_shared` reflects its long-tailed jackpot distribution rather than reliable optimisation.

fixed points — together roughly 32% of all 3,000 runs), can reflect differences in *how often* an algorithm falls into a degenerate basin as much as differences in genuine optimisation quality. Statistical significance here should not be read directly as a difference in trading skill.

D. Why Training Fitness Did Not Predict Test Fitness

Train–test rank correlation (Spearman's ρ on per-run fitness pairs, both initialisation modes pooled, $n = 600$ per strategy) reveals a structural pattern (Figure 5). `sma` ($d = 2$) shows the strongest transfer, $\rho = 0.66$ ($p = 1.9 \times 10^{-77}$): an optimiser that ranks high on training tends to rank high on test. `ema_independent` ($d = 4$) follows at $\rho = 0.38$ ($p = 3.6 \times 10^{-22}$), `weighted` ($d = 14$) at $\rho = 0.34$ ($p = 2.3 \times 10^{-17}$), and `lma` ($d = 2$) at $\rho = 0.31$ ($p = 7.3 \times 10^{-15}$) — all moderate but statistically reliable. `ema_shared` ($d = 3$) shows no detectable transfer at all ($\rho = 0.05$, $p = 0.20$, n.s.).

The relationship is not monotonic in dimensionality. `ema_shared` ($d = 3$) has the *worst* transfer of any strategy, worse than the $d = 14$ `weighted` space; and the only structural difference between `ema_shared` and `ema_independent` is a single parameter — decoupling the two EMAs' smoothing rates — yet that one change moves ρ from 0.05 to 0.38. This indicates that the *nature* of the parameterisation matters more than its size: a tightly-coupled $d = 3$ space can yield markedly less generalisable optima than a decoupled $d = 4$ space or even a $d = 14$ weighted combination.

TABLE VI. Summary of *test* fitness by algorithm and strategy, pooled across both initialisation modes (100 runs per cell). Compare against Table IV: training medians above $100,000 on `ema_shared` collapse to a test median of $1,000 for every algorithm.

Algo	Strategy	Mean	Std	CV	Median	Max
abo	ema_independent	1,455.93	800.27	0.550	1,239.42	4,998.46
aos	ema_independent	1,079.86	496.95	0.460	1,104.87	4,249.85
gps	ema_independent	2,633.91	1,703.55	0.647	1,982.11	5,850.78
mrfo	ema_independent	1,000.00	0.00	0.000	1,000.00	1,000.00
pso	ema_independent	3,288.24	1,485.39	0.452	4,435.27	4,565.80
sos	ema_independent	3,806.42	1,374.95	0.361	4,565.80	5,159.96
abo	ema_shared	1,903.99	1,636.50	0.860	1,000.00	8,795.49
aos	ema_shared	1,914.91	1,653.88	0.864	1,000.00	7,284.67
gps	ema_shared	1,970.13	2,187.39	1.110	1,000.00	9,614.94
mrfo	ema_shared	1,868.67	1,771.49	0.948	1,000.00	7,660.07
pso	ema_shared	2,265.04	1,993.23	0.880	1,000.00	8,392.40
sos	ema_shared	2,190.03	2,061.31	0.941	1,000.00	8,663.88
abo	lma	718.72	186.57	0.260	671.51	907.98
aos	lma	799.00	173.74	0.217	907.98	977.77
gps	lma	906.20	217.94	0.241	931.86	1,399.39
mrfo	lma	759.18	197.79	0.261	907.98	907.98
pso	lma	659.46	174.85	0.265	671.51	907.98
sos	lma	907.98	0.00	0.000	907.98	907.98
abo	sma	1,170.00	22.46	0.019	1,180.28	1,180.28
aos	sma	1,165.29	13.35	0.011	1,153.51	1,180.28
gps	sma	1,080.49	248.38	0.230	1,055.30	1,857.14
mrfo	sma	1,180.28	0.00	0.000	1,180.28	1,180.28
pso	sma	1,112.79	155.32	0.140	1,180.28	1,180.28
sos	sma	1,180.28	0.00	0.000	1,180.28	1,180.28
abo	weighted	4,453.79	1,124.67	0.253	4,620.62	6,498.67
aos	weighted	4,606.32	831.97	0.181	4,745.43	5,920.21
gps	weighted	2,302.15	1,651.62	0.717	1,891.33	5,610.75
mrfo	weighted	4,570.30	977.47	0.214	4,755.59	7,604.38
pso	weighted	4,510.11	993.36	0.220	4,766.00	5,965.44
sos	weighted	4,840.99	516.32	0.107	4,749.80	5,889.75

The result is ironic. The two strategies with the most baseline-beating test runs — `ema_shared` (34 wins) and `weighted` (26 wins) — are exactly the two with the weakest train–test transfer ($\rho = 0.05$ and $\rho = 0.34$), while `sma`, which has the strongest transfer ($\rho = 0.66$), produced zero wins across 600 runs. Win rate in this experiment is, if anything, *anti*-correlated with the reliability of optimisation. The `ema_shared` wins in particular cannot be credited to any optimiser: with $\rho = 0.05$, training performance carries no information about test outcome, so a winning run is a draw from a long right tail rather than the product of effective search. The strategies where training rank *does* predict test rank (`sma`, `lma`, `ema_independent`) are predictably mediocre on test: their best-training algorithm is also their best-test algorithm, but neither beats the baseline. Higher expressiveness produced much higher training fitness yet transferred little or no useful information to held-out test data.

Table VII exposes the failure of generalisation directly. The single highest test result in the entire study — $9,615, achieved by four separate GPS `ema_shared` runs — came from a training fitness of only $16,290, an order of magnitude below the training fitness of runs that tested far lower (SOS reached $166,428 in training yet tested at $8,664). That four distinct GPS runs converged to *exactly* $16,290.48 in training and *exactly* $9,614.94 on test indicates a flat region of the fitness landscape: multiple parameter settings yield identical fitness, so training fitness cannot discriminate among them. Selecting bots by training fitness would therefore not have selected the bots that generalised best — and on `ema_shared` it would have actively selected worse ones.

E. Which Algorithm Performs Best

The question of which optimiser performs best does not have a clean answer in this setting, and the more defensible reading is that the experiment cannot rank optimisers on out-of-sample trading performance at all. With only 2.07%

FIG. 4. Dunn's post-hoc pairwise p-values (Bonferroni-corrected) for all strategy × initialisation combinations — test fitness.

of all runs exceeding the passive baseline, and with those wins concentrated on the two strategies where training fitness does not transfer, there is no reliable test-side signal to separate the optimisers on the metric that matters.

What can be stated holds with care. On `ema_independent` test fitness, SOS and PSO are the strongest performers (SOS pooled median \$4,566, CV = 0.361; PSO pooled median \$4,435, CV = 0.452), while AOS collapses despite leading the same strategy on *training* fitness. On `weighted` test fitness, PSO, SOS, AOS, and MRFO show comparable medians near \$4,750. No single algorithm achieves both high training fitness and reliable test transfer. The apparent stability of MRFO and SOS on `sma` and `lma` (CV = 0.000) reflects convergence to broken fixed points, not robustness — zero variance there indicates collapse. A further pattern worth noting is that GPS, the deliberately weak single-state baseline, exhibits the smallest train-to-test gap of any algorithm. This is not a generalisation advantage: GPS simply never reaches the high training fitness that the population methods do, so it has little fitness to lose on transfer. Its small gap reflects weak optimisation on both splits rather than robust optimisation on either.

This strategy-dependence of relative algorithm performance is consistent with the No Free Lunch theorem [9]: no algorithm dominates uniformly across fitness landscapes. In this experiment the landscape structure — set by the

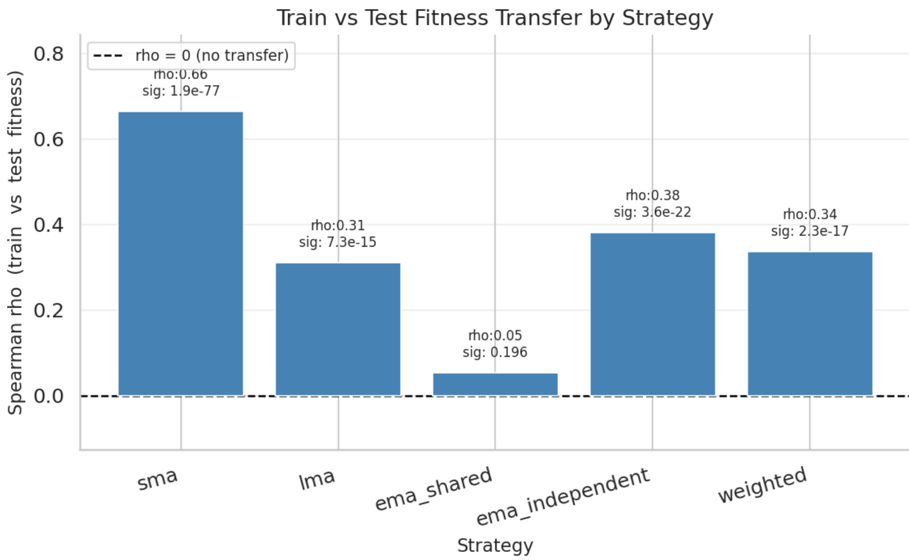

FIG. 5. Spearman rank correlation between training and test fitness per strategy (both initialisation modes pooled, $n = 600$ per strategy). $\rho = 0$ corresponds to no rank transfer.

TABLE VII. The 20 highest test-fitness runs across all 3,000 trials. Training fitness is shown alongside for comparison: the ordering by test fitness bears little relation to the ordering by training fitness.

Algo	**Strategy**	**Start**	**Train Cash**	**Test Cash**
gps	ema_shared	fixed	16,290.48	9,614.94
gps	ema_shared	fixed	16,290.48	9,614.94
gps	ema_shared	random	16,290.48	9,614.94
gps	ema_shared	random	16,290.48	9,614.94
abo	ema_shared	random	54,301.41	8,795.49
sos	ema_shared	fixed	166,428.24	8,663.88
sos	ema_shared	fixed	128,331.33	8,392.40
sos	ema_shared	fixed	114,909.60	8,392.40
gps	ema_shared	fixed	47,972.89	8,392.40
pso	ema_shared	random	138,076.13	8,392.40
sos	ema_shared	random	87,164.40	8,392.40
gps	ema_shared	random	30,519.05	8,392.40
pso	ema_shared	fixed	80,647.43	8,267.02
mrfo	ema_shared	random	125,386.17	7,660.07
mrfo	weighted	random	48,030.39	7,604.38
mrfo	ema_shared	fixed	105,631.68	7,586.17
abo	ema_shared	random	71,590.32	7,499.73
aos	ema_shared	random	41,659.30	7,284.67
pso	ema_shared	random	115,538.41	7,226.59
pso	ema_shared	fixed	107,480.63	7,185.07

choice of strategy — accounts for far more of the variation in outcomes than the choice of optimiser does.

F. Interpretation

The near-universal failure of training fitness to transfer to test is best attributed to the evaluation setup rather than to any optimiser. The decisive evidence is that the train-to-test fitness gap scales with an algorithm's *optimisation success*, not with its identity: GPS, which reaches a pooled training median far below the population methods, has a train/test ratio near 1.0 and does not overfit, whereas SOS, which reaches the highest training fitness, shows the largest gap. Within `ema_shared`, training medians span from \$7,900 (GPS) to \$128,000 (SOS) while every algorithm's test median sits at exactly $1,000 (Table IV versus Table VI). An optimiser maximising the stated objective therefore buys no test improvement. This is the signature of a misspecified objective being optimised faithfully, not of a defective optimiser.

Two non-exclusive mechanisms underlie this. The first is overfitting to the training market regime: the fitness function rewards raw final cash over a single continuous pass of the pre-2020 BTC series, with no penalty for path-specificity, trade count, or drawdown, leaving the optimiser free to fit parameter sets to the particular price path of 2014–2019. The COVID-19 crash, the 2021 bull run, and the 2022 bear market introduce qualitatively different dynamics that such parameters are not positioned to exploit; the pre-2020/post-2020 split is, by construction, an out-of-distribution stress test that punishes exactly this memorisation. The second is that MA-crossover signals may be fundamentally weak predictors of BTC price movement across regimes regardless of parameter tuning: if strategy expressiveness is the binding constraint, no optimiser can recover transferable advantage from the parameter space. The 2.07% rate of bots exceeding buy-and-hold across 3,000 trials is consistent with sampling from the right tail of a high-variance distribution and does not constitute evidence of a learnable trading edge under the MA-crossover hypothesis space — a result consistent with, though not a formal test of, the efficient market hypothesis for short-horizon technical signals on highly liquid assets.

These two mechanisms cannot be separated with the present data, because doing so requires a counterfactual the current design does not include: the same optimisers run against a risk-adjusted objective (for example a Sharpe ratio [10] or return-per-trade metric) under the same train/test split. If the train–test gap narrows under such an objective, the raw-cash fitness function was the dominant cause; if it persists, market non-stationarity dominates. Distinguishing the two is left as the primary item of future work.

VI. CONCLUSION

Algorithm choice has a universally significant effect on training fitness across all strategy and initialisation conditions, yet this advantage does not transfer to test performance. The binding constraint appears to be strategy expressiveness in the face of market non-stationarity rather than optimiser quality: once the target regime shifts, the parameters optimised on training data no longer generalise regardless of how effectively they were found.

AOS performs comparably to PSO despite lacking personal best memory. On this multimodal fitness landscape, global best guidance via nucleus tracking combined with shell-based population distribution appears sufficient; the additional per-particle history that PSO maintains provides no statistically significant advantage. This is a finding specific to the problem structure — the MA crossover landscape may not present the kind of elongated, directional ridges where personal memory typically helps.

GPS confirms that population diversity is necessary. Single-state local search is consistently and significantly worse than every population-based method across all conditions. The MA parameter landscape contains enough local optima that a deterministic pattern search from a fixed starting point cannot reliably reach high-fitness regions, establishing population-based global search as a structural requirement rather than an optional improvement.

SOS's three-evaluation-per-iteration cost is not justified by its performance differential over PSO. Because Dunn's test establishes distributional equivalence between PSO and SOS on training fitness under matched-iteration budgets, the threefold evaluation overhead buys no statistical advantage. A compute-normalised comparison equalising total fitness evaluations rather than iterations would be required to confirm this quantitatively, but the direction of the finding is clear.

Independent α in `ema_independent` provides genuinely greater expressiveness than the shared-α `ema_shared` configuration. The expanded search space produces more pronounced inter-algorithm differences (higher Kruskal-Wallis H-statistics), indicating that the additional degrees of freedom challenge the optimisers' exploration capabilities in a way that better reveals their structural differences.

The LE degradation in AOS is an empirically observed phenomenon. The lowest-energy electron's role as a local-optima escape mechanism degrades as the nucleus shifts across iterations and other electrons improve, because the LE designation tracks relative ranking within the current population rather than an absolute landscape property. This is a genuine weakness in the paper's formulation that warrants further investigation.

The no-trade bad optimum is an unpenalised artifact of using raw final cash as the fitness metric. A risk-adjusted metric such as the Sharpe ratio or return-per-trade would penalise conservative non-trading behaviour and remove this fixed point from the feasible optima.

Outlook Future work should adopt risk-adjusted fitness functions and regime-aware train/test splits that better reflect the non-stationarity of cryptocurrency markets. The train/test divergence observed here suggests the evaluation objective itself may require redesign: objectives that penalise in-sample overfitting — for example, through cross-validated segment fitness or walk-forward testing — would impose generalisation pressure that the current segment-average fitness does not provide.

[1] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proceedings of ICNN'95 – International Conference on Neural Networks*, vol. 4, pp. 1942–1948, IEEE, 1995.
[2] Y. Shi and R. Eberhart, "A modified particle swarm optimizer," in *Proceedings of the IEEE International Conference on Evolutionary Computation*, pp. 69–73, IEEE, 1998.
[3] M. Azizi, "Atomic orbital search: A novel metaheuristic algorithm," *Applied Mathematical Modelling*, vol. 93, pp. 657–683, 2021.
[4] J. B. Odili, M. N. M. Kahar, and S. Anwar, "African Buffalo optimization: A swarm-intelligence technique," *Procedia Computer Science*, vol. 76, pp. 443–448, 2015.
[5] W. Zhao, Z. Zhang, and L. Wang, "Manta ray foraging optimization: An effective bio-inspired optimizer for engineering applications," *Engineering Applications of Artificial Intelligence*, vol. 87, p. 103300, 2020.
[6] M.-Y. Cheng and D. Prayogo, "Symbiotic Organisms Search: A new metaheuristic optimization algorithm," *Computers and Structures*, vol. 132, pp. 189–213, 2014.
[7] V. Torczon, "On the convergence of pattern search algorithms," *SIAM Journal on Optimization*, vol. 7, no. 1, pp. 1–25, 1997.
[8] Investopedia Team, "What are the most common periods used in creating moving average (ma) lines?," 2026.
[9] D. H. Wolpert and W. G. Macready, "No free lunch theorems for optimization," *IEEE transactions on evolutionary computation*, vol. 1, no. 1, pp. 67–82, 1997.
[10] Investopedia Team, "Sharpe ratio: Formula, definition, and how to use it," 2026.